

Концентрация меры на сфере

Homework

1 сентября 2026 года

1. Prove that $|B_2^n| \rightarrow 0$ as $n \rightarrow \infty$, that almost all of the volume of B_2^n lies within $O(1/n)$ of the boundary, and that almost all of it also lies within $O(1/\sqrt{n})$ of any fixed central hyperplane.
2. (IMC) Find the maximum number of points on the unit sphere of $\mathbb{R}^n$ with pairwise distances strictly greater than $\sqrt{2}$.
3. Prove the Gaussian concentration of the norm: for a standard Gaussian vector X in $\mathbb{R}^n$, $\mathbb{P}(|\|X\| - \sqrt{n}| > t) \leq 2e^{-t^2/2}$.
4. (Johnson–Lindenstrauss 1984) Prove that for any N points in $\mathbb{R}^n$ and $\varepsilon \in (0, 1)$ there is a linear map into $\mathbb{R}^k$, $k = O(\varepsilon^{-2} \log N)$, changing all pairwise distances by a factor at most $1 \pm \varepsilon$.
5. (MIPT 2016) Find the volume of the set of points of the cube $[0, 1]^n$ whose coordinates pairwise differ by more than $d > 0$.
6.
 - a) Prove that for two independent uniform points of $[0, 1]^n$ the distance divided by $\sqrt{n}$ converges in probability to $1/\sqrt{6}$.
 - b) Prove that two independent uniform points of S^{n-1} are nearly orthogonal: the angle between them lies in $(\pi/2 - \varepsilon, \pi/2 + \varepsilon)$ with probability at least $1 - 2e^{-n\varepsilon^2/2}$; deduce that one can find e^{cn} points on the sphere that are pairwise nearly orthogonal.
7. State Dvoretzky's theorem and prove the following case: the cube $[-1, 1]^n$ has a central section of dimension $c \log n$ which is 2-Euclidean.
8. Prove that the median and the mean of a 1-Lipschitz function on S^{n-1} differ by at most $C/\sqrt{n}$, so that Lévy's lemma holds with the mean in place of the median.

9. (Hoeffding 1963) Prove that a function on $\{0, 1\}^n$ which is 1-Lipschitz with respect to the Hamming distance satisfies $\mathbb{P}(|f - \mathbb{E}f| > t) \leq 2e^{-2t^2/n}$ for the uniform measure.
10. (Proposed problems, №361) 88% of the surface of a sphere is coloured red. Prove that there is a cube inscribed in the sphere all of whose vertices are red.
11. Prove that the expected maximum of n independent standard Gaussians is $(1 + o(1))\sqrt{2\log n}$, and explain why $\log N$ is the right order in Johnson–Lindenstrauss.
12. Prove that the unit sphere S^{n-1} admits an ε -net of cardinality at most $(1 + 2/\varepsilon)^n$, and that any ε -separated set has at most this many points; deduce the standard 9^n bound used in operator-norm estimates.